

A disproof of Conjecture 00000008189

A Poisson(1/2) irregularity index forces density $1 - \frac{13}{8}e^{-1/2}$ for index ≥ 3

1 The conjecture

Conjecture 00000008189 reads (English):

Conjecture: The density of irr_pairs (the frequency of irregular primes p) is $1 - e^{-1/2}$; the irregularity index of p (the number of zeros) is Poisson(1/2) (a half-Poisson law); the density of primes with large irregularity index (at least 3) is $e^{-1/2}(1/2)^3/3!$ (an explicit sparsity); and the expected number of counterexamples in the next exponential extension of the verified threshold 10^9 of the Vandiver conjecture (to 10^{11}) is 0.00.

The Chinese text gives the same third clause with “ ≥ 3 ”. Write the mathematical clauses as (A) irregular primes have density $1 - e^{-1/2}$; (B) the index is Poisson(1/2); (C) primes of index at least 3 have density $e^{-1/2}(1/2)^3/3!$. The fourth clause (an expected count of 0.00) is not a mathematical assertion about the primes, and it is not formalized. We prove that (B) and (C) are inconsistent, so the conjunction is false.

2 Definitions

- Irregularity index (standard; Buhler–Harvey, <https://arxiv.org/abs/0912.2121v2>: “The index of irregularity of p ...is the number of such irregular indices k ”): $i(p) = \#\{k \text{ even}, 2 \leq k \leq p-3 : p \mid \text{num}(B_k)\}$, with B_k Mathlib’s `bernoulli` (Lean `irregIndex`). Only even $k \geq 2$ occur, so the sign convention for B_1 is irrelevant.
- Relative natural density among the primes: a property P has density d if $\#\{p \leq X \text{ prime} : P(p)\} / \#\{p \leq X \text{ prime}\} \rightarrow d$ as $X \rightarrow \infty$ (Lean `HasPrimeDensity`).
- (B) is read as: for every $k \geq 0$, the primes with $i(p) = k$ have density $\text{Po}(1/2)\{k\} = e^{-1/2}(1/2)^k/k!$ (Lean `HalfPoissonLaw`, using Mathlib’s `ProbabilityTheory.poissonMeasure` and `poissonMeasure_real_singleton`). For an $\mathbb{N}$ -valued statistic this is the same as convergence of the empirical distribution to Poisson(1/2).
- (C) is read literally: the primes with $i(p) \geq 3$ have density $e^{-1/2}(1/2)^3/3!$ (Lean `LargeIndexClause`); (A): the primes with $i(p) \geq 1$ have density $1 - e^{-1/2}$ (Lean `IrregularClause`).

3 The proof

Lemma 1 (Lean `card_partition`, `ratio_sum`). *For $X \geq 2$ the four ratios for $i = 0, i = 1, i = 2, i \geq 3$ sum to 1.*

Proof. The four classes partition the primes $\leq X$, and there is at least one such prime. $\square$

Lemma 2 (Lean `tail_ne`). *If $\text{Po}(1/2)\{0\} + \text{Po}(1/2)\{1\} + \text{Po}(1/2)\{2\} + d_3 = 1$ then $d_3 \neq e^{-1/2}(1/2)^3/3!$.*

Proof. The hypothesis says $d_3 = 1 - (1 + \frac{1}{2} + \frac{1}{8})e^{-1/2} = 1 - \frac{13}{8}e^{-1/2}$. If also $d_3 = e^{-1/2}/48$ then $\frac{79}{48}e^{-1/2} = 1$, so $e^{-1/2} = 48/79$ and $e = (79/48)^2 = 6241/2304 < 2.71$. But $e > 2.7182818283$ (Mathlib `Real.exp_one_gt_d9`). $\square$

Numerically the forced value is $1 - \frac{13}{8}e^{-1/2} \approx 0.014388$ and the stated value is $e^{-1/2}/48 \approx 0.012636$.

Theorem 3 (Lean `halfPoisson_contradicts_largeIndex`, `conjecture_8189_false`). $\neg(B \wedge C)$, and hence $\neg(A \wedge B \wedge C)$.

Proof. Under (B) and (C) the four ratios converge to $\text{Po}(1/2)\{0\}, \text{Po}(1/2)\{1\}, \text{Po}(1/2)\{2\}$ and $e^{-1/2}/48$. Their sum is eventually the constant 1, so by uniqueness of limits the four limits sum to 1, contradicting the previous lemma. $\square$

Remark 4. Only finite additivity and normalization are used, so the same contradiction holds for any finitely additive density on the primes (logarithmic, Dirichlet); `tail_ne` is stated for arbitrary d_3 . No property of Bernoulli numbers is used, and nothing is claimed about whether (A) or (B) is true (both are open heuristics of Lehmer and Siegel; Buhler–Harvey: “This is motivated by the heuristic (Lehmer, Siegel) that if the Bernoulli numerators are random modulo $p \dots$ ”).

4 Reading not refuted

If “(at least 3)” / “ ≥ 3 ” were a slip for “exactly 3”, then (C) would be the $k = 3$ case of (B), the clauses would be consistent, and nothing here would refute the conjecture. We do not adopt that reading: both languages say at least 3, and “large irregularity index” names a threshold, not a single value. The reviewer should weigh this; the value $e^{-1/2}(1/2)^3/3!$ is exactly $\text{Po}(1/2)\{3\}$.

5 Lean formalization

File `lean/Conjecture8189/Basic.lean` (119 lines; only import `Mathlib`). Main theorems `C8189.halfPoisson_contradicts_largeIndex` : $\neg(\text{HalfPoissonLaw} \wedge \text{LargeIndexClause})$ and `C8189.conjecture_8189_false` : $\neg(\text{IrregularClause} \wedge \text{HalfPoissonLaw} \wedge \text{LargeIndexClause})$. `lake build` succeeds; `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. No sorry, no `native_decide`, no new axioms.